

1 HI-Jass User Manual

1.1 Running the application

1.1.1 Linux

`./run_hi_jass.sh`

1.1.2 Windows

Double-click `run_hi_jass.bat`, or run:

`.\run_hi_jass.bat`

The launcher creates the virtual environment, installs the dependencies, and starts the GUI.

1.2 Default NBI configuration

The default configuration contains two neutral beams:

Beam	Species	Injection energy	Injected power	Tangent R_t	Direction
NBI-1	D	120 keV	10 MW	0.5 m	co
NBI-2	T	180 keV	0.1 MW	0.6 m	counter

1.3 Two-beam slowing-down model

The slowing-down quantities are calculated independently for each beam. For beam j :

$$E_{c,j} = \left(\frac{m_j}{m_e} \right)^{1/3} T_e$$

$$\tau_{se,j} = 6.27 \times 10^8 \frac{A_j T_e [\text{eV}]^{3/2}}{n_e [\text{cm}^{-3}] \ln \Lambda}$$

$$\tau_{S,j} = \frac{\tau_{se,j}}{3} \ln \left[1 + \left(\frac{E_{b,j}}{E_{c,j}} \right)^{3/2} \right]$$

The useful beam power includes the HotJass capture and loss cascade. The fast-ion density from each beam is:

$$n_{b0,j} = \frac{P_{\text{useful},j} \tau_{S,j}}{E_{b,j} (10^3 e) V}$$

The total fast-ion density is:

$$n_{b0} = \sum_j n_{b0,j}$$

1.4 Density peaking and shine-through

The density peaking input defines the analytic electron-density profile:

$$n_e(\rho) = n_{e0}(1 - \rho^2)^{2p_n}, \quad 0 \leq \rho \leq 1$$

Shine-through uses the density along the beam chord rather than assuming that the central density applies everywhere. Each NBI supplies a tangent point (R_t, Z_t) in the NBI tab. The straight tangential chord at that point is intersected with the shaped plasma cross-section, and the optical depth is evaluated as:

$$\tau_b = \sigma_{\text{stop}}(E_b/A_b) \int_{\text{chord}} n_e(\rho(s)) ds$$

and the captured and shine-through fractions are:

$$f_{\text{capt},b} = 1 - e^{-\tau_b}, \quad f_{\text{shine},b} = e^{-\tau_b}.$$

The chord is evaluated numerically using the local cylindrical radius along the tangent path and the plasma elongation. With $p_n = 0$, the density is uniform along the computed chord; with $p_n > 0$, the profile is integrated point by point. A tangent point outside the plasma produces zero captured power.

1.4.1 Shine-through model selection

Each NBI has a three-way shine-through selector:

- **Riviere:** the existing Riviere-type analytic stopping cross-section fit.
- **Janev/Suzuki:** the Janev-Boley-Post analytic fit from the supplied paper, Nuclear Fusion 29 (1989) 2125, Eqs. (23)-(26), using the published Table III coefficients and the carbon-impurity coefficients from Table IV. It models total effective stopping, including the paper's density, electron-temperature and Z_{eff} dependence. The fit is published for $100 \leq E \leq 10^4$ keV/u, $10^{12} \leq n_e \leq 10^{15}$ cm⁻³ and $1 \leq T_e \leq 50$ keV; inputs outside these ranges are clipped to the nearest validity boundary.
- **Manual:** the entered manual shine-through fraction is used directly; captured power is $(1 - f_{\text{shine}})P_{NB}$.

The selected calculated stopping model is also used by the optional first-orbit-loss attenuation weighting. The Manual mode bypasses the cross-section calculation for shine-through.

1.5 First-orbit loss (reduced model)

First-orbit loss is an **opt-in** channel (the *First-orbit loss* checkbox in the Models section). When enabled, it estimates the fraction of *captured* fast ions that leave the plasma promptly on their first drift orbit, before any slowing-down or charge-exchange because they are born close enough to the last closed flux surface (LCFS) that their guiding-centre orbit, or their gyro-orbit, crosses it.

The **Orbit model** selector (Models section) chooses how the orbit widths and the loss criterion are built:

- **Large-aspect** ($q_* \rho_{Li}$) the original reduced model described in *Passing-orbit width* and *Loss criterion* below. Direction-only mean shift; exactly zero co-current loss.
- **ST orbits mean-shift (arbitrary A)** and **ST orbits pitch-resolved** two spherical-tokamak variants described in *ST orbit models* at the end of this section. Both give non-zero co-current loss.

1.5.1 Birth profile along the chord

The calculation reuses the shine-through deposition. Along the straight tangential chord (assumed $R_t = R_0$, length $2a$) the local ionisation rate density is

$$\dot{n}(x) \propto n_{e0} \sigma_{\text{stop}}(E_{b,j}/A_j) e^{-n_{e0} \sigma_{\text{stop}} x}, \quad x \in [0, 2a],$$

and the path coordinate is mapped to a normalised flux label by

$$\rho(x) = \frac{|x - a|}{a},$$

so $\rho = 0$ at the chord mid-point and $\rho = 1$ at both ends. The stopping cross-section is the beam's selected shine-through model; **Manual** mode has no deposition shape, so the Riviere fit is substituted for this weighting only.

Off-axis / vertically-shifted injection. When a beam's tangency radius R_t differs from R_0 , or its aim height $Z_t \neq 0$, the birth profile is built instead from the *real* chord: a horizontal line tangent to the cylinder $R = R_t$ at height Z_t , sampled through elliptical flux surfaces

$$\rho(s)^2 = \left(\frac{R(s) - R_0}{a} \right)^2 + \left(\frac{Z_t}{\kappa a} \right)^2, \quad R(s) = \sqrt{R_t^2 + (s - s_0)^2},$$

with the chord length running edge-to-edge and ρ no longer reaching 0 (the beam no longer passes through the magnetic axis). This is the same geometry the shine-through / capture calculation already uses. For the standard on-axis tangential case ($Z_t = 0$, $R_t = R_0$ or unset) the crude $\rho(x) = |x - a|/a$ mapping above is kept, so existing results are unchanged.

Central column. The beam enters from the outboard side. If its tangency radius is inside the central column ($R_t < R_{\text{post}}$, set by **Centre-post** R; default $R_0 - a$), the straight chord would strike the column, so only the **outboard leg** from the outboard-edge entry to the point where $R(s) = R_{\text{post}}$ is kept; the (unreachable) far leg is discarded. A beam with R_t at or outboard of R_{post} threads the gap between the column and the inboard plasma edge and keeps its full chord. A beam whose aim point lies outside the plasma altogether contributes no first-orbit loss. The Deposition tab's geometry panel draws the column and marks a blocked beam.

1.5.2 Passing-orbit width

The radial width of a strongly co-passing drift orbit at the full injection energy is the textbook large-aspect-ratio estimate

$$\Delta r_j = q_* \rho_{Li,j}, \quad q_* = \frac{2\pi a^2 B_t}{\mu_0 R_0 I_p} \frac{1 + \kappa^2}{2}, \quad \rho_{Li,j} = \frac{m_j v_{b,j}}{e B_t}.$$

q_* is the cylindrical, elongation-corrected safety factor and $\rho_{Li,j}$ the toroidal-field gyroradius at $E_{b,j}$.

1.5.3 Loss criterion

The orbit is modelled as a **rigid mean shift** of the birth radius: inward for co-current injection (favourable), outward for counter-current:

$$\rho_{\text{eff}}(x) = \rho(x) \mp \frac{\Delta r_j}{a} \quad (\text{co / counter}),$$

and an ion born at x is counted as lost if $\rho_{\text{eff}}(x) > 1$. Because $\rho(x) \leq 1$, the co-current shift can never trigger this test, so the mean-shift model gives **exactly zero first-orbit loss for co-current injection**; for counter-current it loses every ion born with $\rho(x) > 1 - \Delta r_j/a$.

An optional, direction-independent **Larmor (gyro-orbit) prompt-loss** channel can be OR-ed in: an ion whose guiding centre is born within one gyro-diameter of the LCFS,

$$\rho(x) > 1 - \frac{2\rho_{Li,j}}{a},$$

is lost regardless of the drift direction (gyration is much faster than the guiding-centre drift). This channel is disabled by default.

1.5.4 Loss fraction and power

$$f_{\text{orbit},j} = \frac{\int_{\text{lost}} \dot{n}(x) dx}{\int_0^{2a} \dot{n}(x) dx}, \quad P_{\text{orbit},j} = f_{\text{orbit},j} P_{\text{capt},j}.$$

First-orbit loss is applied to the captured power *before* the charge-exchange loss and before the useful power that drives T_e , T_i and $n_{b0,j}$. Injection direction (co-/counter-current) is set per NBI in its input section.

1.5.5 Reported quantities

The Dashboard and the Assumptions tab report, per beam, $\rho_{Li,j}$ and Δr_j (in cm and as fractions of a) and $f_{\text{orbit},j}$.

1.5.6 Limitations

- **The co-current zero is a model artefact, not physics.** The rigid mean-shift ignores the finite radial width of the orbit and the outboard excursion that a favourable inward mean shift does not cancel; published orbit-following comparisons give of order a few percent co-current first-orbit loss, versus tens of percent counter-current.
- The chord label $\rho(x) = |x - a|/a$ folds the inboard and outboard halves together, so the model cannot distinguish a high-field-side birth from a low-field-side birth. It therefore cannot represent the correct structure co-current losing high-field-side edge births, counter-current losing low-field-side edge births, and trapped-ion banana-tip losses on the low-field side in either direction.
- Circular / elliptical cross-section, large-aspect-ratio orbit widths (the birth chord itself now honours R_t and Z_t , but the orbit-width and pitch estimates are still evaluated at R_0); Δr_j and $\rho_{Li,j}$ are evaluated at the full injection energy (an upper bound real ions slow down).
- The birth-profile integral in this channel uses the flat central density n_{e0} , not the peaked profile used elsewhere.

- When ρ_{Li}/a or $\Delta r/a \gtrsim 0.3$ orbit width comparable to the machine size, typical of compact low-aspect-ratio devices the whole estimate is order-unity uncertain and really needs a guiding-centre orbit follower. The Assumptions tab flags this case. The ST orbit models below address the first three of these limitations directly.

1.5.7 ST orbit models (arbitrary aspect ratio)

On a spherical tokamak the drift and banana orbit widths are an $O(a)$ fraction of the minor radius, the trapped fraction is large, and outboard-born trapped ions are lost **regardless of injection direction** so the large-aspect model's exact co-current zero is qualitatively wrong. The two ST variants build their widths from the **poloidal gyroradius**

$$\rho_{\theta,j} = \frac{B_t}{\bar{B}_p} \rho_{Li,j}, \quad \bar{B}_p = \frac{\mu_0 I_p}{2\pi a \sqrt{(1 + \kappa^2)/2}},$$

(not $q_* \rho_{Li}$, whose q_* runs away as $\varepsilon \rightarrow 1$), together with the circulating and banana widths and the arbitrary- A trapped fraction

$$w_{\text{pass},j} = \varepsilon \rho_{\theta,j}, \quad w_{\text{ban},j} = 2\sqrt{\varepsilon} \rho_{\theta,j}, \quad f_t = 1 - \frac{(1 - \varepsilon)^2}{(1 + 1.46\sqrt{\varepsilon})\sqrt{1 - \varepsilon^2}},$$

with $\varepsilon = a/R_0$ (clamped to $[10^{-3}, 0.95]$) and $f_c = 1 - f_t$ (Lin-Liu & Miller, *Phys. Plasmas* **2** (1995) 1666). An arbitrary- A edge safety factor q_a (Uckan / ITER Physics Basis form, with the $(1.17 - 0.65\varepsilon)/(1 - \varepsilon^2)^2$ shaping factor) is reported for reference but is **not** used in the widths.

The birth profile in this channel uses the same Beer-Lambert deposition density as the large-aspect model and the Deposition tab's chord plot, $n_e \sigma e^{-n_e \sigma x}$ via the selected stopping model so it responds to n_{e0} the same way everywhere in the code (an earlier version used a fixed "penetration rule of thumb" mean free path that saturated at its clamp for realistic densities, decoupling the ST models' loss fraction from n_{e0} ; this has been fixed).

The **drift-orbit shift is direction-dependent** for both populations inward for co-current ($v_{\parallel}$ along I_p), outward for counter with a weaker shift $c_t w_{\text{pass}}$ ($c_t = 0.25$) for the trapped-ion banana centre than for a passing ion. The **gyro channel** ($\rho > 1 - \rho_{Li}/a$, born within one gyroradius of the LCFS) is direction-independent. Co-current loss is therefore non-zero but smaller than counter-current the co/counter asymmetry survives, unlike the large-aspect model's all-or-nothing 0 vs 1.

Mean-shift variant. Deposition-weighted loss integrals over $\rho(x)$, blended by f_c, f_t (sign convention: $-$ co, $+$ counter):

$$f_{\text{orbit},j} = f_c L_{\text{pass}} + f_t L_{\text{trap}},$$

$$L_{\text{pass}} : \rho \mp \frac{w_{\text{pass},j}}{a} > 1 \quad \text{or} \quad \rho > 1 - \frac{\rho_{Li,j}}{a},$$

$$L_{\text{trap}} : \rho + \frac{w_{\text{ban},j}}{2a} \mp \frac{0.25 w_{\text{pass},j}}{a} > 1 \quad \text{or} \quad \rho > 1 - \frac{\rho_{Li,j}}{a}.$$

Pitch-resolved variant. A 2-D integral over birth radius x and the pitch **magnitude** $|\lambda| = |v_{\parallel}/v|$. The birth value $|\lambda_0(x)| = R_t/\sqrt{R_t^2 + (x - a)^2}$ is fixed by the injection *geometry* and does **not** depend on the poloidal-field / current direction only the sign of $v_{\parallel}$ relative to I_p flips (co

$\leftrightarrow$ counter), carried by $s_{\text{dir}} = \mp 1$. $|\lambda_0|$ is smeared by a $\sigma = 0.10$ Gaussian. An ion is trapped if $|\lambda| < \sqrt{2\varepsilon_{\text{loc}}/(1 + \varepsilon_{\text{loc}})}$ ($\varepsilon_{\text{loc}} = \rho a/R_0$); its outboard radial excursion is

$$\text{passing : } s_{\text{dir}} \frac{w_{\text{pass},j}}{a} |\lambda|, \quad \text{trapped : } \frac{w_{\text{ban},j}}{2a} \frac{|\lambda|}{\lambda_{\text{tr}}} + s_{\text{dir}} \frac{0.25 w_{\text{pass},j}}{a},$$

(the banana half-width $\rightarrow 0$ for deeply trapped ions, largest near the trapped/passing boundary), and it is lost if $\rho + (\text{excursion}) > 1$ or $\rho > 1 - \rho_{Li,j}/a$.

Both variants collapse toward the large-aspect co-current zero as $\varepsilon \rightarrow 0$. They remain 0-D order-of-magnitude estimates—absolute magnitudes are upper-bound-like and, because the direction-independent trapped banana-tip and gyro channels stay significant, the co/counter split is narrower than the large-aspect model’s (typically a factor ~ 1.5 , not 0 vs 1). References: Akers *et al.*, *Nucl. Fusion* (START NBI, $A \sim 1.4$); Goldston, White & Boozer, *Phys. Rev. Lett.* **47** (1981) 1004; Goldston & Rutherford, *Introduction to Plasma Physics* (1995), Ch. 12.

1.6 Charge-exchange loss

A captured fast ion can charge-exchange with a background cold neutral during slowing-down, becoming a fast **neutral** that escapes the plasma before fully thermalizing—a second loss channel, applied to the post-orbit-loss captured power (Step 0, same ordering as first-orbit loss above). Design note: docs/CX_model.pdf. A **CX-loss model** selector (rail, "Models" section) picks between three modes; only the active mode’s inputs matter.

1.6.1 Manual fraction (default)

The original flat efficiency knob `cx_loss_fraction` $\in [0, 1]$: a fixed fraction of captured power is removed, independent of E_b , n_e , or any background-neutral density. PPPL-1280 Sect. 6.2.2/6.2.3 quotes "up to 10%" of injected power for $Z_{\text{eff}} = 1$ as a rough qualitative figure, not a formula. Default 0=off; leaves every other result byte-identical.

1.6.2 Manual n_0/n_e the physics-based survival integral

The user supplies n_0/n_e directly (a spatially uniform, 0-D background neutral density $n_0 = (n_0/n_e)n_{e0}$), and the model evaluates the actual CX survival integral along the slowing-down path. For a fast ion at energy E :

$$\nu_{\text{cx}}(E) = n_0 \sigma_{\text{cx}}(E) v(E), \quad S(E) = \exp \left[- \int_E^{E_b} \frac{\nu_{\text{cx}}(E')}{|dE'/dt|} dE' \right],$$

with particle- and power-loss fractions

$$f_{\text{cx},N} = 1 - S(0^+), \quad f_{\text{cx},P} = \int_0^{E_b} \frac{E}{E_b} \frac{\nu_{\text{cx}}(E)}{|dE/dt|} S(E) dE.$$

$f_{\text{cx},P}$ (the quantity actually applied to P_{useful}) is smaller than $f_{\text{cx},N}$ because σ_{cx} falls steeply above ~ 3050 keV/amu, so most CX events happen late in the cascade, when little energy is left to remove. $1/|dE/dt| = (\tau_{se}/2)\sqrt{E}/(E^{3/2} + E_c^{3/2})$ is exactly the same slowing-down kernel already used by `thermalization_time/slowing_down_distribution` (differentiating $\tau_s(E) = (\tau_{se}/3) \ln[1 + (E/E_c)^{3/2}]$ reproduces it), so no new slowing-down physics is introduced—only the extra $\nu_{\text{cx}}(E)$ factor under the same integral. $\sigma_{\text{cx}}(E)$ is the Janev & Smith (1993, Sect. 2.3.1)

analytic form, refit by Swaczyna, Bzowski & Kubiak (arXiv:2411.13174, 2024, their Eq. A1) against Schultz *et al.* (2023) theory and the Barnett (1990) tabulation quoted accuracy 5% from 1100 keV/amu, the range this project’s beams live in.

Also reported, per beam, on the Dashboard: the dimensionless figure of merit $\gamma_{\text{cx}} \equiv \nu_{\text{cx}}(E_b) \tau_s \approx n_0 \sigma_{\text{cx}}(E_b) v_b \tau_s$ negligible for $\gamma_{\text{cx}} \ll 1$, worth taking seriously for $\gamma_{\text{cx}} \sim 0.11$, and a sign the single-pass loss treatment is breaking down for $\gamma_{\text{cx}} \gtrsim 1$ (flagged with a warning, as is $f_{\text{cx},P} > 10\%$, PPPL-1280’s rough $Z_{\text{eff}} = 1$ ceiling). Because CX loss depends on T_e (via $E_c(T_e)$) while T_e itself depends on the CX-reduced P_{useful} , this is a genuine fixed point; an outer under-relaxed Picard loop (in `hotjass_core`, mirroring the existing alpha-heating fixed point) closes it, keeping the core solver a single non-iterative solve.

1.6.3 Penetration (n0_LCFS/ne) edge-penetration estimate

Instead of a single uniform n_0 , the boundary value $n_{0,\text{LCFS}} = (n_{0,\text{LCFS}}/n_e) n_{e,\text{LCFS}}$ is set from the user ratio, and the core profile follows the standard penetration/recycling picture:

$$\lambda_0(\rho) = \frac{v_0}{n_e(\rho) [\langle \sigma v \rangle_{\text{ion}}(T_e) + \langle \sigma v \rangle_{\text{cx}}(T_i)]}, \quad n_0(\rho) \approx n_{0,\text{LCFS}} \exp \left[- \int_{\rho}^1 \frac{a}{\lambda_0(\rho')} d\rho' \right],$$

volume-averaged to a single $\langle n_0 \rangle$ and fed through the same survival integral as ”Manual n0/ne” above. v_0 is a round ~ 3 eV Franck-Condon-like dissociation speed. **Caveat:** the ionization ($\langle \sigma v \rangle_{\text{ion}}(T_e)$) and thermal-CX ($\langle \sigma v \rangle_{\text{cx}}(T_i)$) rate coefficients used for λ_0 are an interim hand-built approximation (a threshold-shaped form for ionization, a threshold-free $\sqrt{T}$ form for CX) a calibrated Voronov (1997) / ADAS `adf11` fit was not sourced/verified in this session, so treat $n_0(\rho)$ ’s absolute scale as order-of-magnitude only; the survival-integral physics downstream of $\langle n_0 \rangle$ (same $\sigma_{\text{cx}}(E)$ fit as ”Manual n0/ne”) is not affected by this caveat.

1.6.4 Escape probability (informational only)

A fast neutral born from a CX event can re-ionize before reaching the wall, so treating every CX event as a full loss is a conservative upper bound, not exact. An escape-probability factor $\exp(-x_{\text{wall}}/\lambda_{\text{reion}})$ is shown per beam (Dashboard) $x_{\text{wall}} \approx a$, λ_{reion} from the same combined ionization+CX stopping cross section a beam neutral itself uses but is **not** applied to $f_{\text{cx},P}$; it is informational, per the design note’s explicit ”acceptable as the conservative default, but shown as such.”

1.7 Power balance (electron and ion)

At each central density the model solves a 0-D steady-state power balance for T_e and T_i . The heating that drives each temperature is the sum of the neutral-beam split, prescribed auxiliary heating (ECRH and ICRH), and when enabled fusion alpha self-heating.

1.7.1 Beam heating split

Each useful beam j delivers a fraction $L_e(x_j)$ of its power to electrons and $L_i(x_j) = 1 - L_e(x_j)$ to ions, with $x_j = E_{b,j}/E_{c,j}(T_e)$ and

$$L_i(x) = \frac{1}{x} \int_0^x \frac{du}{1 + u^{3/2}}, \quad L_e(x) = 1 - L_i(x).$$

$L_i \rightarrow 1$ as $x \rightarrow 0$ (cold, ion-dominated) and $L_i \rightarrow 0$ as $x \rightarrow \infty$ (fast beam, electron-dominated). $P_{\text{useful},j}$ is the injected power after shine-through, first-orbit loss and charge-exchange loss (preceding sections).

1.7.2 Auxiliary heating (ECRH, ICRH)

ECRH and ICRH are entered as powers with electron/ion fractions. Their contribution to each channel is

$$P_{\text{aux},e} = f_e^{\text{EC}} P_{\text{ECRH}} + f_e^{\text{IC}} P_{\text{ICRH}}, \quad P_{\text{aux},i} = (1 - f_e^{\text{EC}}) P_{\text{ECRH}} + f_i^{\text{IC}} P_{\text{ICRH}}.$$

ECRH is electron-dominant, f_e^{EC} close to 1. For ICRH the electron and ion fractions are set independently; if $f_e^{\text{IC}} + f_i^{\text{IC}} < 1$ the remainder is treated as power that does not thermalise (a fast-ion tail or direct loss). These sources are Te/Ti-independent constants added to the right-hand side of each balance.

1.7.3 Alpha self-heating (optional)

When enabled, 3.5 MeV fusion alphas deposit

$$P_\alpha = f_\alpha \frac{E_\alpha}{E_f} P_{f,\text{tot}} = f_\alpha \frac{3.5}{17.6} P_{f,\text{tot}},$$

split between electrons and ions by the alpha slowing-down fraction

$$L_e^\alpha = L_e \left(\frac{E_\alpha}{E_c^\alpha(T_e)} \right), \quad E_c^\alpha = \left(\frac{m_\alpha}{m_e} \right)^{1/3} T_e,$$

so $P_{\alpha,e} = L_e^\alpha P_\alpha$ and $P_{\alpha,i} = (1 - L_e^\alpha) P_\alpha$. Because P_α depends on $P_{f,\text{tot}}$, which depends on T_i , which depends on $P_{\alpha,i}$, the alpha term is closed by an under-relaxed fixed-point iteration; with alpha heating off the solve is a single pass.

1.7.4 Decoupled balance (default)

With electron-ion equipartition off, the two balances are solved in sequence. The electron balance is a 1-D root-find for T_e :

$$\frac{3}{2} n_{e0} T_e (10^3 e) \frac{V}{\tau_{E,e}} = \sum_j L_e(x_j) P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e}.$$

The ion temperature then follows in closed form (or, for the neoclassical-ion option, from a second 1-D root-find see *Confinement time*):

$$\frac{3}{2} n_{\text{thermal}} T_i (10^3 e) \frac{V}{\tau_{E,i}} = \sum_j L_i(x_j) P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i},$$

where $n_{\text{thermal}} = n_{\text{sum}}(n_{e0}, Z_{\text{eff}}) - n_{b0}$ is the thermal-ion density left after the fast ions are subtracted. A non-positive n_{thermal} marks the operating point infeasible.

1.7.5 Coupled balance (equipartition on)

When electron-ion equipartition is enabled, the exchange power $P_{ei}(T_e, T_i)$ (defined in *Electron-ion exchange time*) is subtracted from the electron balance and added to the ion balance, and the pair is solved together:

$$\frac{3}{2}n_{e0}T_e(10^3e)\frac{V}{\tau_{E,e}} = \sum_j L_e(x_j)P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e} - P_{ei}(T_e, T_i),$$

$$\frac{3}{2}n_{\text{thermal}}T_i(10^3e)\frac{V}{\tau_{E,i}} = \sum_j L_i(x_j)P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i} + P_{ei}(T_e, T_i).$$

The reported P_e and P_i remain the pure beam split; P_{ei} , P_{α} and P_{aux} are reported separately.

1.7.6 Profile-corrected 0-D

The balance above is genuinely 0-D: one density, one T_e , one T_i , treated as uniform over the volume. With the **profile-corrected 0-D** checkbox (Models section) *off* the default the entered central density is used directly as that uniform value and the solved T is a flat-plasma effective temperature; comparing it to a measured on-axis T_0 then needs a peaking factor supplied by the reader (validation runs use $\simeq 2$).

With the checkbox *on*, the entered **central n.e** is taken to be the **on-axis** n_{e0} , and for a $(1-\rho^2)^{2p}$ profile the volume average is $\langle X \rangle = X_0/(1+2p)$. The energy balance and the τ_E scalings then run on

$$\langle n_e \rangle = \frac{n_{e0}}{1+2p_n},$$

so the solved T_e, T_i are **volume-averaged**. The electron and ion temperature profiles are shaped by **independent** exponents p_{Te} (**Temp peaking (electron)**) and p_{Ti} (**Temp peaking (ion)**); a value < 0 means "use p_{Te} "), so the on-axis values are

$$T_{e0} = \langle T_e \rangle (1+2p_{Te}), \quad T_{i0} = \langle T_i \rangle (1+2p_{Ti}).$$

They are reported (Dashboard row "on-axis T_{e0}/T_{i0} ") and are what the fusion, pressure and β_t integrals use as their central values the reactivity integrals with the $(1-\rho^2)^{2p_{Ti}}$ ion shape, the beam slowing-down kernel with the $(1-\rho^2)^{2p_{Te}}$ electron shape. Set p_n (**Density peaking**), p_{Te} and p_{Ti} to match the real profiles; $p \simeq 0.5075$ gives $X_0/\langle X \rangle \simeq 22.5$ (L-/H-mode), while a **hot-ion mode** on a compact ST needs $p_{Ti} \gg p_{Te}$ ($T_{i0}/\langle T_i \rangle \simeq 34$ with T_e nearly flat). With all peaking parameters at 0 the correction is a no-op.

1.8 Confinement time

$\tau_{E,e}$ and $\tau_{E,i}$ in the balances above are set per channel by the *Confinement* selector in the Models section; the electron and ion channels are chosen independently. When a physics-based scaling is selected the manual **tauE,e** / **tauE,i** inputs are ignored, and the Summary sheet reports the computed value (or its range across a scan) instead.

The physics-based scalings below all use the same loss power

$$P_{\text{loss}} = \sum_j P_{\text{useful},j} + P_{\text{aux},e} + P_{\text{aux},i},$$

i.e. the captured, post-orbit, post-charge-exchange beam power plus the prescribed auxiliary (ECRH, ICRH) heating. Fusion alpha heating is not included in P_{loss} (it is closed by an outer fixed-point and is not known when τ_E is evaluated). None of these depend on T_e or T_i , so each is evaluated once.

1.8.1 Fixed (input)

$\tau_{E,e}$ and $\tau_{E,i}$ are the values entered on the Plasma tab, used unchanged.

1.8.2 IPB98(y,2) ELMy H-mode

The ITER Physics Basis ELMy H-mode scaling (ITER Physics Basis, *Nucl. Fusion* **39** (1999) 2175; standard reference form):

$$\tau_E = 0.0562 I_p^{0.93} B_t^{0.15} n_{19}^{0.41} P_{\text{loss}}^{-0.69} R_0^{1.97} \kappa_a^{0.78} \varepsilon^{0.58} M_{\text{eff}}^{0.19}$$

with I_p in MA, B_t in T, P_{loss} in MW, R_0 and a in m, $n_{19} = n_{e0}/10^{19} \text{ m}^{-3}$, $\varepsilon = a/R_0$, $\kappa_a = \kappa$, and $M_{\text{eff}} = 2x_D + 3x_T$ the mass-weighted ion mass number. Selecting *IPB98(y,2) ELMy H-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$); *IPB98(y,2) e / neoclassical i* uses it for the electron channel only, with the ion channel from the neoclassical estimate below.

The fit was obtained from conventional-aspect-ratio devices ($A \sim 2.5$ -4) and is documented in the spherical-tokamak literature to mis-predict at low aspect ratio. When $A = R_0/a < 2$ the Assumptions tab flags that IPB98(y,2) is being extrapolated outside its dataset.

1.8.3 Kaye NSTX L-mode

An NSTX low-aspect-ratio (spherical tokamak) L-mode global fit (Kaye, *Nucl. Fusion* **46** (2006) 848, Eq. 5 / Fig. 9):

$$\tau_E = 4.73 \times 10^{-4} I_p^{1.01} B_t^{0.70} n_e^{-0.07} P_{\text{loss}}^{-0.37}$$

with I_p in A, B_t in T, n_e in m^{-3} , P_{loss} in W and τ_E in s. Selecting *Kaye NSTX L-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$); *Kaye L e / neoclassical i* uses it for the electron channel only.

1.8.4 Kaye NSTX H-mode

The H-mode counterpart of the previous fit: an NSTX low-aspect-ratio (spherical tokamak) H-mode thermal-energy scaling (Kaye, *Nucl. Fusion* **46** (2006) 848, Table 1, ordinary-least-squares "Case 1" fit to all 85 H-mode points, RMSE = 0.145):

$$\tau_E = 4.69 \times 10^{-9} I_p^{0.57} B_t^{1.08} n_e^{0.44} P_{\text{loss}}^{-0.73}$$

with I_p in A, B_t in T, n_e in m^{-3} , P_{loss} in W and τ_E in s. Relative to IPB98(y,2) the current dependence is much weaker ($I_p^{0.57}$ vs $I_p^{0.93}$) and the toroidal-field dependence much stronger ($B_t^{1.08}$ vs $B_t^{0.15}$), the qualitative spherical-tokamak trend; the density and heating-power exponents

$(n_e^{0.44}, P_{\text{loss}}^{-0.73})$ are of the same order as the conventional-aspect scalings. This is an ST-appropriate fit ($A \sim 1.31.5$ dataset). Selecting *Kaye NSTX H-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$); *Kaye H e / neoclassical i* uses it for the electron channel only.

1.8.5 Neoclassical ion

A banana-regime order-of-magnitude estimate (Wesson, *Tokamaks*; Chang-Hinton low-collisionality limit) for the ion channel only:

$$\chi_{i,\text{neo}} \simeq \frac{q^2 \rho_i^2 \nu_{ii}}{\varepsilon^{3/2}}, \quad \tau_{E,i} = \frac{a^2}{\chi_{i,\text{neo}}} = \frac{a^2 \varepsilon^{3/2}}{q^2 \rho_i^2 \nu_{ii}},$$

with $\varepsilon = a/R_0$, the elongation-corrected cylindrical safety factor

$$q = \frac{5 a^2 B_t}{R_0 I_p [\text{MA}]} \frac{1 + \kappa^2}{2},$$

the thermal-ion Larmor radius

$$\rho_i = \frac{m_{\text{eff}} v_{th,i}}{e B_t}, \quad v_{th,i} = \sqrt{\frac{2 T_i (10^3 e)}{m_{\text{eff}}}}, \quad m_{\text{eff}} = x_D m_D + x_T m_T,$$

and the ion-ion collision frequency summed over D and T,

$$\nu_{ii} = \sum_{s \in \{D, T\}} 4.80 \times 10^{-8} \frac{n_s [\text{cm}^{-3}] \ln \Lambda}{\sqrt{\mu_s} T_i [\text{eV}]^{3/2}}, \quad \mu_s = m_s / m_p.$$

Since $\tau_{E,i} \propto T_i^{1/2}$, the ion balance becomes a 1-D fixed-point root-find for T_i rather than a closed-form substitution. Pure neoclassical transport has no anomalous channel, so it under-estimates real ion transport and can predict very hot, sometimes implausible T_i ; treat it as a lower bound. It cannot be combined with electron-ion equipartition.

1.8.6 Neoclassical ion (arbitrary aspect ratio)

The bare $\varepsilon^{-3/2}$ geometric factor above both diverges as $\varepsilon \rightarrow 0$ and stays bounded as $\varepsilon \rightarrow 1$, neither of which is right on a spherical tokamak. The *arbitrary A* variant (selector entries ending "neoclassical i (arbitrary A)") replaces the trapped-fraction part with the Lin-Liu & Miller f_t/f_c the banana-regime structure of Helander & Sigmar's *Collisional Transport in Magnetized Plasmas*, Ch. 11 and uses the arbitrary- A edge safety factor q_a :

$$\chi_{i,\text{neo}} \simeq q_a^2 \rho_i^2 \nu_{ii} \frac{f_t}{f_c}, \quad f_t = 1 - \frac{(1 - \varepsilon)^2}{(1 + 1.46\sqrt{\varepsilon})\sqrt{1 - \varepsilon^2}}, \quad f_c = 1 - f_t,$$

$$q_a = \frac{5 a^2 B_t}{R_0 I_p [\text{MA}]} \frac{1 + \kappa^2 (1 + 2\delta^2 - 1.2\delta^3)}{2} \frac{1.17 - 0.65 \varepsilon}{(1 - \varepsilon^2)^2}.$$

f_t/f_c stays finite for every $\varepsilon < 1$ and rises steeply only as $\varepsilon \rightarrow 1$ (all ions trapped) the physically correct low-aspect trend. This is **not** a smooth reduction of the $\varepsilon^{-3/2}$ form: the two order-of-magnitude estimates can differ by up to about one order of magnitude in either direction (more once the $\tau_{E,i} \propto T_i^{1/2}$ fixed point amplifies it), which is exactly why both are offered run them

side by side. References: Helander, *Phys. Plasmas* **7** (2000) 3999; Hinton, Wiley *et al.*, *Phys. Rev. Lett.* **29** (1972) 698; Satake *et al.*, *Phys. Plasmas* **9** (2002); Goldston & Rutherford (1995).

1.9 Mean fast-ion energy

Each beam has its own steady-state slowing-down distribution:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \quad 0 < E \leq E_{b,j}$$

Its mean fast-ion energy is:

$$\langle E_{fast} \rangle_j = \frac{\int_0^{E_{b,j}} E f_j(E) dE}{\int_0^{E_{b,j}} f_j(E) dE}$$

1.10 Combined values shown in Results

The GUI reports power-weighted effective values for the two beams.

Effective slowing-down time:

$$\tau_S^{\text{eff}} = \frac{\sum_j P_{\text{useful},j} \tau_{S,j}}{\sum_j P_{\text{useful},j}}$$

Effective mean fast-ion energy:

$$\langle E_{fast} \rangle = \frac{\sum_j P_{\text{useful},j} \langle E_{fast} \rangle_j}{\sum_j P_{\text{useful},j}}$$

These effective values are used for the combined fast-ion diagnostics, including fast-ion energy density, pressure, beta, and:

$$R = \frac{u_{fast}}{U_t}$$

The underlying HotJass calculation still evaluates each beam separately before forming these combined diagnostics.

1.11 Results tabs (Operating point)

- **Dashboard** scalar table of the operating point (temperatures, and their on-axis values when profile-corrected 0-D is on; densities, power balance terms, orbit widths, D-T / D-D fusion power, neutron rate, Greenwald ratio, edge safety factor q_* , beta). q_* is reported using whichever formula the active orbit-loss model actually uses – the bare cylindrical `safety_factor_cyl_edge` for "large-aspect", or the arbitrary- A `safety_factor_cyl_edge_arbitrary_A` (q_a) for either ST orbit variant – so it stays consistent with the orbit widths shown just above it. Per-beam CX-loss rows show $f_{\text{cx},P}/f_{\text{cx},N}/\gamma_{\text{cx}}$ (plus n_0 and the informational escape probability) when a physics-based CX model is selected, or just the flat fraction for "Manual fraction" – see "Charge-exchange loss" above.

- **Deposition** (1) beam targeting geometry in the torus top view (titled with the active device), tangent to each beam's R_t with its co-/counter-current sense; (2) neutral-beam survival $I(s)/I_0$ and the fast-ion birth rate $n_e \sigma I(s)/I_0$ along the beam, annotated with each beam's shine-through percentage; (3) the resulting fast-ion birth versus normalised radius ρ , per beam and total (chord-sampled, midplane circular approximation), with the prompt first-orbit-loss zone $\rho > \rho_{cut}$ shaded and a per-beam ρ_{cut} line (from the same criteria as the orbit-loss model in use); (4) the steady-state slowing-down distribution $f(E)$ with the E_b edge marked.
- **Power flow** waterfall / Sankey / pie of injected power to its sinks.
- **Profiles** $n_e(\rho)$, $T_{e,i}(\rho)$, the plasma shape, $\tau_S(\rho)$ (local thermalization time, from the same local $n_e(\rho)$, $T_e(\rho)$ used elsewhere in this panel – log scale, since it rises sharply toward the edge as $n_e \rightarrow 0$), and $P_{fus}(\rho)$ (local D-T + D-D fusion power density, thermal vs. beam-plasma / beam-target, built from the same on-axis densities and $(1 - \rho^2)^{2p}$ shapes as the volume-integrated totals on the Dashboard – see "Fusion power" below).
- **Assumptions** the full validity / fit-range read-out and any warnings.
- **References** the literature behind the *currently selected* models (beam stopping, confinement scaling, orbit-loss model) and the active machine geometry, each with a link, plus a feedback button.

1.12 Electron-ion exchange time

The electron-ion exchange time is based on the NRL Plasma Formulary thermal-equilibration rate. For each thermal ion species s (D or T):

$$\bar{\nu}_{ei,s} = 1.8 \times 10^{-19} \frac{\sqrt{m_e m_s} Z_e^2 Z_s^2 n_e \ln \Lambda}{(m_s T_e + m_e T_i)^{3/2}}$$

In this practical-unit expression, masses are in grams, temperatures are in eV, and electron density is in cm^{-3} . The model uses $Z_e = Z_s = 1$ for hydrogenic D and T ions.

For a D-T plasma, the effective exchange time is:

$$IE = \frac{n_D + n_T}{n_D \bar{\nu}_{ei,D} + n_T \bar{\nu}_{ei,T}}$$

For a single thermal ion species, this reduces to:

$$IE = \frac{1}{\bar{\nu}_{ei}}$$

The equivalent energy-over-power form used by the application is:

$$IE = \frac{\frac{3}{2}(n_D + n_T)|T_i - T_e|(10^3 e)V}{|P_{ie}|}$$

where V is the plasma volume and $e = 1.602176634 \times 10^{-19}$ J/eV. The exchange power is:

$$P_{ie} = -P_{ei} = -\frac{3}{2}V(T_e - T_i)(n_D \bar{\nu}_{ei,D} + n_T \bar{\nu}_{ei,T})(10^3 e)$$

The absolute values make τ_{IE} positive regardless of whether electrons heat ions ($T_e > T_i$) or ions heat electrons ($T_i > T_e$).

1.13 Fusion power

The total fusion power is the sum of the thermal D-T and beam-target contributions:

$$P_f = P_{f,\text{thermal}} + P_{f,\text{beam}}$$

1.13.1 Thermal D-T fusion

The thermal contribution is calculated from the thermal deuterium and tritium densities and the ion-temperature-dependent Bosch-Hale reactivity:

$$P_{f,\text{thermal}} = n_{D0}n_{T0}\langle\sigma v\rangle_{DT}(T_i)VE_f$$

Here n_{D0} and n_{T0} are the thermal species densities, V is the plasma volume, and:

$$E_f = 17.6 \text{ MeV} = 17.6 \times 10^6 e$$

The Bosch-Hale thermal reactivity is evaluated at T_i , because the relative velocity is that of the thermal ions, not the electrons.

1.13.2 Beam-target fusion

Each useful beam is evaluated separately against the stationary thermal ions of the other D-T species:

$$P_{f,\text{beam}} = \sum_j VE_f \int_0^{E_{b,j}} n_{\text{target},j} f_j(E) [\sigma v]_j(E) dE$$

The beam slowing-down distribution is normalized to the fast-ion density of that beam:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \quad \int_0^{E_{b,j}} f_j(E) dE = n_{b0,j}$$

The beam-target reactivity is:

$$[\sigma v]_j(E) = \sigma_{DT}(E)v_j(E), \quad v_j(E) = \sqrt{\frac{2E}{m_j}}$$

For a D-T beam-target reaction $E_f = 17.6 \text{ MeV}$. For a D beam, $n_{\text{target},j} = n_{T0}$; for a T beam, $n_{\text{target},j} = n_{D0}$. The beam fast-ion density is obtained from the useful beam power:

$$n_{b0,j} = \frac{P_{\text{useful},j} TS_{j,j}}{E_{b,j}(10^3 e)V}$$

The input shine-through fraction is used to determine captured beam power; the plotted P_{shine} is then the derived shine-through power from the scan. Charge-exchange and other configured losses similarly reduce the useful power before calculating $n_{b0,j}$.

The fusion post-processing always uses the $(1 - \rho^2)$ profile weighting: the thermal contribution is

$$P_{f,\text{thermal}} = V E_f \langle n_D(\rho) n_T(\rho) \langle \sigma v \rangle_{DT} [T_i(\rho)] \rangle_V,$$

and the beam-target contribution uses the local target density and the local slowing-down distribution evaluated with $T_e(\rho)$. The n_{D0} , n_{T0} , T_e , T_i used here are the **on-axis** values (see *Profile-corrected 0-D*).

1.13.3 D-D fusion

The D-D channel (present whenever the deuterium fraction $x_D > 0$) is evaluated with the Bosch-Hale D-D fits, **branch-resolved**:

branch	products	E_f	neutron
D(d,p)T	T (1.01) + p (3.02 MeV)	4.033 MeV	none
D(d,n) $\bar{\text{s}}$ He	$\bar{\text{s}}$ He (0.82) + n (2.45 MeV)	3.269 MeV	2.45 MeV

Thermal:

$$P_{f,\text{DD}}^{\text{th}} = V [E_{ddp} \langle \frac{1}{2} n_D^2 \langle \sigma v \rangle_{ddp}(T_i) \rangle_V + E_{ddn} \langle \frac{1}{2} n_D^2 \langle \sigma v \rangle_{ddn}(T_i) \rangle_V],$$

with the $\frac{1}{2}$ identical-particle factor and the same $(1 - \rho^2)$ profile weighting as the D-T thermal term.

Beam-target (a fast D ion on the thermal-D population **no** $\frac{1}{2}$ factor, the two populations are distinct):

$$P_{f,\text{DD}}^{\text{beam}} = \sum_{j \in \text{D beams}} V \int_0^{E_{b,j}} n_{D0} f_j(E) [E_{ddp} \sigma_{ddp}(E) + E_{ddn} \sigma_{ddn}(E)] v_j(E) dE.$$

The cross-sections take the **lab-frame** deuteron energy (same convention as σ_{DT}). D-D does **not** produce alphas, so the alpha self-heating fixed-point closes on the D-T power only ($P_\alpha = f_\alpha(3.5/17.6)P_{f,DT}$).

1.13.4 Neutron rate

$$R_n = \frac{P_{f,DT}}{17.6 \text{ MeV}} + R_{ddn}^{\text{th}} + R_{ddn}^{\text{beam}},$$

i.e. one 14.06 MeV neutron per D-T reaction plus the 2.45 MeV neutrons from the D(d,n) branch. Reported on the Dashboard and as **R_neutron** in a scan.

1.13.5 Current model limitations

- The beam-target integral uses stationary target ions, so the solved T_i is not included in that reaction integral, and the beam is treated as a single full-energy component (no half/third fractions).
- The tritium and $\bar{\text{s}}$ He bred by D-D are not burned further ("cat-DD" is neglected).
- The radial volume weighting uses the model's elliptical flux-surface approximation and does not yet solve a 2D equilibrium.

1.14 Density scan

The Plasma tab defines the requested scan range with `n_e_min` and `n_e_max`. The Results tab reports the physically valid interval separately. Points below the HotJass charge-neutrality feasibility limit are not assigned physical output values and are omitted from plots.

The T_e, T_i scan plot's y-axis top is $\min(\max(T_e, T_i), 100 \text{ keV})$ – the actual data maximum, unless it exceeds 100 keV (the upper edge of the Bosch-Hale fits' validated range), in which case the axis is capped there so one runaway hot-ion point doesn't wash out the rest of the curve. The bottom is fixed at 0 (not autoscaled) so a fixed top never stretches the axis into an apparent negative margin.

The n_D, n_T, n_{b0} scan group's otherwise-empty 4th panel shows the fast-ion / target-ion density ratio n_{b0}/n_{Target} , where n_{Target} is whichever thermal D-T species the *dominant* (by injected power) NBI beam reacts with in the beam-target channel – n_T for a D beam, n_D for a T beam.

1.15 Important notation

- T_e : electron temperature
- T_i : thermal-ion temperature
- P_e : beam power deposited into electrons
- P_i : beam power deposited into ions
- L_e, L_i : beam-power electron / ion heating fractions ($L_e + L_i = 1$)
- P_{ie} : electron-ion energy exchange power
- $P_{\text{ECRH}}, P_{\text{ICRH}}$: ECRH and ICRH auxiliary heating power
- $P_{\text{aux},e}, P_{\text{aux},i}$: auxiliary heating to electrons / ions
- $P_\alpha, P_{\alpha,e}, P_{\alpha,i}$: fusion alpha self-heating power (total, to electrons, to ions)
- $\tau_{E,e}, \tau_{E,i}$: electron / ion energy confinement times used in the balance
- n_{thermal} : thermal-ion density, $n_{\text{sum}}(n_{e0}, Z_{\text{eff}}) - n_{b0}$
- n_D, n_T : thermal deuterium and tritium densities
- n_{b0} : total fast-ion density
- ρ_{Li} : fast-ion toroidal-field Larmor radius at the injection energy
- Δr : passing drift-orbit radial width, $q_* \rho_{Li}$ (large-aspect model)
- ρ_θ : fast-ion poloidal gyroradius, $(B_t/\bar{B}_p) \rho_{Li}$ (ST orbit models)
- $w_{\text{pass}}, w_{\text{ban}}$: ST circulating / banana orbit widths
- f_t, f_c : trapped / circulating particle fractions (arbitrary A)
- q_a : arbitrary-aspect-ratio edge safety factor (Uckan form)
- f_{orbit} : first-orbit loss fraction of captured beam power
- P_{orbit} : first-orbit loss power
- $P_{f,\text{tot}}, P_{f,\text{th}}, P_{f,b}$: total, thermal, and beam-target fusion power (D-T)
- $P_{f,DT}, P_{f,DD}$: D-T and D-D fusion power ($P_{f,\text{tot}} = P_{f,DT} + P_{f,DD}$)
- R_n : neutron rate (14 MeV D-T + 2.45 MeV D-D branch)
- $\langle T \rangle, T_0$: volume-averaged and on-axis temperature (profile-corrected 0-D)
- p_n, p_{Te}, p_{Ti} : density and electron / ion temperature profile exponents ($(1 - \rho^2)^{2p}$)
- τ_{IE} : effective electron-ion exchange time
- $p_{\text{th}}, p_{\text{fast}}$: thermal and fast-ion pressure
- β_t : toroidal beta